

1.1 Roots of polynomial equations

Further Pure 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

For polynomial equations, Vieta's formulas link symmetric sums/products of roots to coefficients. If coefficients are real, non-real complex roots occur in conjugate pairs. Transformed roots can be found without solving the original polynomial by substituting a new variable and clearing denominators carefully.

Worked example: follow the reasoning

For $x^2 - 5x + 6 = 0$, roots α, β satisfy $\alpha + \beta = 5$, $\alpha\beta = 6$; roots $1/\alpha$ and $1/\beta$ have sum $5/6$ and product $1/6$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A monic cubic has roots 1, 2 and 4. Write equation whose roots are their reciprocals. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Reciprocal roots 1, $\frac{1}{2}$, $\frac{1}{4}$ [1]; multiply $(x-1)(2x-1)(4x-1)$ [1]; $8x^3 - 14x^2 + 7x - 1 = 0$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.